This Patient Group Direction (PGD) is intended only for registered healthcare professionals who have been named and authorised by their organisation to practice under it.

Patient Group Direction

for the treatment of Vitamin B12 Deficiency and Folate Deficiency

Hydroxocobalamin injection, Cyanocobalamin tablets and Folic acid tablets

Investigation before treatment

Patient Group Direction, version 006, issued 10 September 2026. Supersedes Version 005, 10 September 2026.

What to test

- NICE CKS recommends that investigation of suspected vitamin B12 or folate deficiency includes a full blood count, a blood film, serum folate, and either total B12 (serum cobalamin) or active B12 (serum holotranscobalamin).
- Active B12 (holotranscobalamin) is the preferred marker where available. It measures the fraction of B12 available to cells and avoids the falsely low results seen with oral contraceptives and in pregnancy. Where only total B12 is available it remains acceptable, interpreted against the thresholds below.
- A point of care or finger prick device that reports B12 alone does not satisfy this PGD. It may be used to triage or to monitor a patient already diagnosed, but a full blood count, blood film and folate from an accredited laboratory are required before treatment is started under this PGD.
- Where results are indeterminate and symptoms or signs are present, consider serum methylmalonic acid (MMA).
- Ferritin, thyroid function and coeliac serology are reasonable additions where the cause is unclear, but their interpretation is outside this PGD and abnormal results should be referred.

Interpreting the results

Full blood count

- A mean cell volume (MCV) greater than 100 femtolitres indicates macrocytosis. MCV may be normal where there is coexisting iron deficiency, or where anaemia has developed rapidly over a few weeks.
- A normal MCV does not exclude the need for cobalamin testing. Neurological impairment occurs with a normal MCV in around 25% of cases.
- White cell count and platelet count may be reduced where anaemia is severe.
- The reticulocyte count may be low relative to the degree of anaemia, usually 1 to 3%.

Blood film

- Hypersegmented neutrophils (more than 5% of neutrophils with five or more lobes, or one or more neutrophils with six or more lobes) and oval macrocytes may suggest either B12 or folate deficiency, but are neither sensitive nor specific in early cobalamin deficiency.
- Oval macrocytes, hypersegmented neutrophils and circulating megaloblasts are typical features of clinical cobalamin deficiency.

Vitamin B12 level: NICE thresholds

- Confirmed deficiency: total B12 less than 180 nanograms/L (133 picomol/L), or active B12 less than 25 picomol/L.
- Possible deficiency: total B12 180 to 350 nanograms/L (133 to 258 picomol/L), or active B12 25 to 70 picomol/L.
- Unlikely to be deficiency: total B12 more than 350 nanograms/L (258 picomol/L), or active B12 more than 70 picomol/L.

Limitations to hold in mind

- The clinically normal level is unclear. A serum cobalamin below 200 nanograms/L (148 picomol/L) is thought sensitive enough to diagnose 97% of people with deficiency.

- Levels correlate poorly with symptoms. People below 100 nanograms/L (75 picomol/L) usually have clinical or metabolic evidence of deficiency. In older people, levels of 100 to 160 nanograms/L may occur without anaemia or macrocytosis, and clinically significant deficiency can exist with a level in the normal range.
- Women taking oral contraceptives may show reduced cobalamin because of reduced carrier protein, without true deficiency.
- People already taking B12 may have raised total or active levels without the deficiency having been fully treated.
- Serum cobalamin falls in pregnancy and is less reliable there.
- People of Black ethnicity may have a higher reference range for serum B12 than people of White or Asian ethnicity.

Folate level

- Serum folate below 7 nanomol/L (3 micrograms/L) is used as a guide to folate deficiency.
- There is an indeterminate zone at 7 to 10 nanomol/L (3 to 4.5 micrograms/L), so a low folate is suggestive rather than diagnostic.
- Where clinical suspicion is strong but serum folate is normal, red cell folate may be measured once cobalamin deficiency has been excluded. A red cell folate below 340 nanomol/L (150 micrograms/L) is consistent with clinical folate deficiency in the absence of B12 deficiency.

The rule that matters most

Where both B12 and folate are low, B12 must be treated first, or at least at the same time. Giving folic acid alone to someone with untreated B12 deficiency can precipitate or worsen subacute combined degeneration of the spinal cord. If in any doubt about the order of treatment, refer.

Summary of NICE and SmPC guidance for vitamin B12 deficiency treated with hydroxocobalamin injection and folic acid

Vitamin B12 deficiency in over 16s: diagnosis and management, NICE guideline NG239, published 6 March 2024. Hydroxocobalamin 1mg/ml Solution for Injection, Summary of Product Characteristics, ADVANZ Pharma (Mercury Pharmaceuticals Ltd, PL 12762/0008), text revised 23 January 2024, emc updated 27 January 2025. Folic Acid 5mg Tablets, Summary of Product Characteristics, Wockhardt UK Ltd (PL 29831/0097), text revised 4 July 2018. Summarised 10 September 2026.

This summary is part 2 of the house format for a Get Real Health PGD: what the PGD is for, then the guidance that governs the condition, then the PGD itself. It is here so that a pharmacist can hold this document against the guidance it claims to follow. It summarises the source named above and adds nothing to it.

Symptoms and who is at risk

NICE states that the symptoms and signs of vitamin B12 deficiency vary from person to person and are often not exclusive to the deficiency.

Common symptoms and signs listed by NICE are abnormal findings on a blood count such as anaemia or macrocytosis, cognitive difficulties such as difficulty concentrating or short-term memory loss, eyesight problems related to optic nerve dysfunction, glossitis, neurological or mobility problems related to peripheral neuropathy or to central nervous system disease including myelopathy (balance issues and falls, impaired gait, pins and needles or numbness), symptoms of anaemia suggesting iron treatment is not working in pregnancy or breastfeeding, and unexplained fatigue.

NICE says not to rule out the diagnosis solely because anaemia or macrocytosis is absent, and to be aware that the deficiency can be associated with mental health problems including depression, anxiety or psychosis.

Common risk factors listed by NICE are a diet low in vitamin B12 (for example vegan diets or diets excluding animal-source foods, food allergy, difficulty buying or preparing food, low income or a restricted diet), a family history of vitamin B12 deficiency or an autoimmune condition, atrophic gastritis affecting the gastric body, coeliac disease or another autoimmune condition, certain medicines (colchicine, H2-receptor antagonists, metformin, phenobarbital, pregabalin, primidone, proton pump inhibitors and topiramate), previous abdominal or pelvic radiotherapy, previous gastrointestinal surgery including many bariatric operations, gastrectomy or terminal ileal resection, and recreational nitrous oxide use.

NICE states that deficiency is highly likely after total gastrectomy or complete terminal ileal resection if the person is not receiving oral or intramuscular replacement.

NICE recommends offering an initial diagnostic test to people with at least one common symptom or sign and at least one common risk factor, and using clinical judgement for people with a symptom or sign but no risk factor.

How deficiency is confirmed

NICE recommends either total B12 (serum cobalamin) or active B12 (serum holotranscobalamin) as the initial test, except in pregnancy (use active B12) and where recreational nitrous oxide use is the suspected cause (use plasma homocysteine or serum methylmalonic acid).

Blood samples for diagnostic tests should be taken before starting vitamin B12 replacement.

NICE's thresholds for total B12 are: less than 180 nanograms per litre (133 pmol per litre) is confirmed deficiency, between 180 and 350 nanograms per litre (133 to 258 pmol per litre) is an indeterminate result with possible deficiency, and more than 350 nanograms per litre (258 pmol per litre) suggests deficiency is unlikely. The corresponding active B12 thresholds are less than 25 pmol per litre (confirmed deficiency), 25 to 70 pmol per litre (indeterminate) and more than 70 pmol per litre (deficiency unlikely).

NICE says that where there is substantial local variation in validated total B12 thresholds, the thresholds set by the testing laboratory should be used, and that people of Black ethnicity may have a higher reference range for serum vitamin B12.

For symptomatic people with an indeterminate result, NICE says to consider a further test for serum methylmalonic acid, and to consider starting replacement while waiting for that result if the person has a condition that may deteriorate rapidly (for example ataxia or anaemia), an irreversible cause such as autoimmune gastritis, surgery that can cause deficiency, or is pregnant or breastfeeding.

NICE advises caution in interpreting results in people already using over-the-counter vitamin B12 preparations (which may raise levels without fully treating a deficiency) and in people taking the combined oral contraceptive pill (which can lower total B12 without causing a deficiency).

The hydroxocobalamin SmPC states that it is advisable to confirm the diagnosis of vitamin B12 deficiency before giving hydroxocobalamin, and that indiscriminate administration may mask the true diagnosis.

NICE says not to delay replacement while waiting for test results in people with suspected megaloblastic anaemia and neurological symptoms, especially those related to sub-acute combined degeneration of the spinal cord.

Identifying the cause

NICE recommends considering an anti-intrinsic factor antibody test where autoimmune gastritis is suspected, unless the person has previously had a positive test or an operation that could affect absorption.

A negative anti-intrinsic factor antibody result does not rule out autoimmune gastritis, and NICE lists further investigations (anti-gastric parietal cell antibody, gastrin, CobaSorb test, gastroscopy with gastric body biopsy) where suspicion persists.
NICE recommends serological testing for coeliac disease where the cause remains unknown after further investigation.
NICE explains that it does not use the term pernicious anaemia for autoimmune gastritis, because pernicious anaemia in its true sense (life-threatening anaemia) is now extremely rare.

When intramuscular replacement is indicated rather than oral

NICE recommends offering lifelong intramuscular vitamin B12 replacement where autoimmune gastritis is the cause or suspected cause, or the person has had a total gastrectomy or a complete terminal ileal resection.
Where malabsorption has another cause (for example coeliac disease, partial gastrectomy or some forms of bariatric surgery), NICE says to offer replacement and consider intramuscular instead of oral; if oral replacement is used for malabsorption the dosage should be at least 1 mg a day.
For medicine-induced deficiency and for deficiency caused by recreational nitrous oxide use, NICE says to offer either intramuscular or oral replacement based on clinical judgement and the person's preference, and to advise the person to stop using nitrous oxide.
For dietary deficiency NICE says to consider oral replacement, but to consider intramuscular injections instead where the person has another condition that may deteriorate rapidly (for example a neurological or haematological condition), or where there are concerns about adherence (for example older people with multimorbidity or frailty, delirium or cognitive impairment, or social issues such as homelessness).
Where the cause is uncertain and malabsorption is not suspected, NICE says to offer replacement and consider oral instead of intramuscular, reviewing the response at the first follow-up.
If vitamin B12 deficiency is diagnosed in pregnancy or breastfeeding and autoimmune gastritis is suspected, NICE says to start intramuscular replacement without waiting for the antibody test result.

Hydroxocobalamin regimens as stated in the SmPC

NICE does not set out a hydroxocobalamin dosing schedule; recommendation 1.6.11 refers to increasing the frequency of injections in line with the summary of product characteristics, so the regimen below is taken from the SmPC.
The SmPC's licensed indications are Addisonian pernicious anaemia, prophylaxis and treatment of other macrocytic anaemias associated with vitamin B12 deficiency, tobacco amblyopia and Leber's optic atrophy.
For Addisonian pernicious anaemia and other macrocytic anaemias without neurological involvement the SmPC gives initially 250 to 1000 micrograms intramuscularly on alternate days for one or two weeks, then 250 micrograms weekly until the blood count is normal, with maintenance of 1000 micrograms every two to three months.
With neurological involvement the SmPC gives initially 1000 micrograms on alternate days as long as improvement is occurring, with maintenance of 1000 micrograms every two to three months.
For prophylaxis of macrocytic anaemia associated with vitamin B12 deficiency resulting from gastrectomy, some malabsorption syndromes and strict vegetarianism, the SmPC gives 1000 micrograms every two or three months.
The method of administration is intramuscular injection; the only contraindication is hypersensitivity to the active substance or any excipient.
The SmPC states that hydroxocobalamin injection should not be used for the treatment of megaloblastic anaemia of pregnancy, and that it is secreted into breast milk but is unlikely to harm the infant.
The SmPC notes that treatment usually results in rapid haematological improvement, but that neurological symptoms respond more slowly.

Folic acid and the rule about treating B12 first

The folic acid 5 mg SmPC states that folic acid should not be administered for the treatment of pernicious anaemia or undiagnosed megaloblastic anaemia without sufficient amounts of vitamin B12, because folic acid alone will not prevent and may precipitate subacute combined degeneration of the spinal cord, so a full clinical diagnosis should be made before initiating treatment.
The folic acid SmPC adds that folic acid may worsen the symptoms of co-existing vitamin B12 deficiency and should never be used to treat anaemia without a full investigation of the cause.
The hydroxocobalamin SmPC states that if megaloblastic anaemia fails to respond, folate metabolism should be investigated, and that doses in excess of 10 micrograms daily may produce a haematological response in patients with folate deficiency.
The folic acid SmPC dose for folate-deficient megaloblastic anaemia in adults is 5 mg daily for 4 months, with up to 15 mg daily possibly necessary in malabsorption states; the dose for drug-induced folate deficiency is 5 mg daily.
Folic acid is contraindicated in patients with malignant disease unless megaloblastic anaemia is due to folic acid deficiency, and the SmPC advises caution in patients who may have folate-dependent tumours.

Interactions listed in the folic acid SmPC include reduced absorption with sulfasalazine, cholestyramine and aluminium or magnesium antacids, reduced effect with trimethoprim or sulfonamides (which may be serious in megaloblastic anaemia), reduced serum levels of phenytoin, phenobarbital and primidone, and a combination with fluorouracil that should be avoided.
Reported adverse effects of folic acid are rare allergic reactions (including anaphylaxis) and gastrointestinal effects such as abdominal distension, flatulence, anorexia and nausea.

Adverse effects and monitoring of hydroxocobalamin

The hydroxocobalamin SmPC states that regular monitoring of the blood is recommended, and that cardiac arrhythmias secondary to hypokalaemia have been reported during initial therapy, so plasma potassium should be monitored during this period.
Adverse effects listed (all with frequency not known) include hypersensitivity reactions such as rash, itching and exanthema, anaphylaxis, initial hypokalaemia, headache, paraesthesia, tremor, nausea, vomiting, diarrhoea, fever, chills, hot flushes, dizziness, malaise, injection site reactions, acneiform and bullous eruptions, chromaturia, and reactive thrombocytosis during the first weeks of use in megaloblastic anaemia.
The SmPC notes that parenteral chloramphenicol may attenuate the effect of hydroxocobalamin and that oral contraceptives may lower serum hydroxocobalamin concentration, but says these interactions are unlikely to be of clinical significance.

Follow-up, information and referral

NICE recommends explaining that symptoms could start to improve within 2 weeks but this may take up to 3 months, that symptoms may get worse initially before improving, and when to seek medical help if symptoms do not improve, get worse, return or are new.
NICE recommends an initial follow-up at 3 months after starting treatment (earlier depending on severity), or at 1 month if the person is pregnant or breastfeeding.
NICE says not to repeat the initial diagnostic test in people receiving intramuscular replacement.
If symptoms have not sufficiently improved on intramuscular treatment, NICE says to increase the frequency of injections if needed in line with the SmPC, think about alternative diagnoses, and agree a date for reassessment.
For an irreversible cause such as autoimmune gastritis, NICE says to continue lifelong injections even if symptoms have resolved; where a reversible cause has been resolved and symptoms have improved, NICE says to think about stopping or reducing the frequency of injections.
NICE states that people with autoimmune gastritis are at higher risk of gastric neuroendocrine tumours and may be at higher risk of gastric adenocarcinoma, and that new or worsening upper gastrointestinal symptoms should prompt consideration of referral for endoscopy in line with NICE's suspected cancer guideline.
NICE also recommends explaining that people with some causes of deficiency, such as autoimmune gastritis, will need lifelong replacement, and that it is important to continue treatment as advised.

What the sources do not say

NICE NG239 does not give a hydroxocobalamin dose, injection frequency or loading schedule; it refers to the SmPC for this.
NICE NG239 does not state the rule that vitamin B12 must be replaced before or alongside folic acid; that rule comes from the folic acid and hydroxocobalamin SmPCs.
The hydroxocobalamin SmPC does not give a specific number of loading doses for the neurological regimen beyond continuing alternate-day injections as long as improvement is occurring.
The folic acid SmPC does not give a threshold serum or red cell folate value for diagnosing folate deficiency, and NICE NG239 does not cover folate testing.

Patient Group Direction 1 of 3

for the administration of

Hydroxocobalamin 1mg/ml Solution for Injection

for the treatment of vitamin B12 deficiency

Healthcare professionals covered and training

	Healthcare professionals allowed to work under this PGD and requirements
Qualifications and Registration	• Pharmacist registered and practicing with the GPhC • Pharmacy technician registered and practicing with the GPhC
Training and assessment	Pharmacists and Pharmacy Technicians must have completed training relevant to this condition to use this PGD. This training will be documented and overseen by the Get Real Health team. • All users must be familiar with the SPC for the products as well as BNF advice and any applicable national guidelines • All users must have completed training in the interpretation of the blood results relied on by this PGD, covering full blood count, blood film, B12 and folate, and must know when a result requires referral rather than treatment • All users administering by injection must be competent in intramuscular injection technique and in the recognition and management of anaphylaxis • All users must previously have used PGD's to supply medication • Must work in compliance with the SOPs of their own employer • All users must have appropriate indemnity in place to cover this service
Further training and responsibilities	• All users must be up to date with CPD requirements and appraisal • All users must be up to date with MHRA and safety alerts with regards to medical products • All users must understand the concepts of capacity and consent and be aware of the Mental Capacity Act 2005

Clinical condition or situation to which this PGD applies

PGD Indication	Treatment of confirmed vitamin B12 deficiency in adults, including deficiency arising from pernicious anaemia, malabsorption or dietary insufficiency. Hydroxocobalamin supports neurological function, red blood cell formation and DNA synthesis.
Inclusion criteria	• Adults aged 18 years and over. • Confirmed vitamin B12 deficiency on blood testing, interpreted in line with the thresholds above, together with the clinical picture. • A full blood count, blood film and serum folate have been obtained and reviewed alongside the B12 result. • Suitable for intramuscular injection in a non-acute setting. • Valid informed consent obtained. • No contraindication as set out below.
Exclusion criteria	• Known hypersensitivity to hydroxocobalamin or to cobalamin derivatives. • Active infection at the proposed injection site. • Known hereditary problems of cobalamin metabolism.

	- Leber's hereditary optic neuropathy, or a family history of it, because of the risk of optic atrophy. - Severe thrombocytopenia or a bleeding disorder, unless intramuscular injection has been assessed as safe by a clinician familiar with the individual's bleeding risk. - Pregnancy. The SPC states that hydroxocobalamin injection should not be used for the treatment of megaloblastic anaemia of pregnancy, and B12 results are less reliable in pregnancy. Refer to the GP or midwife so that the cause is properly identified and antenatal care is not delayed. - New or progressive neurological symptoms or signs, including sensory disturbance, gait disturbance or cognitive change. Refer for same-week medical assessment; these require prompt investigation and may need a different treatment schedule. - Abnormal full blood count or blood film beyond macrocytic anaemia, including unexplained cytopenias or features suggesting another cause. Refer. - Deficiency not confirmed on testing, or testing incomplete. - Under 18 years of age.
Cautions including any relevant action to be taken	- Where both B12 and folate are low, treat B12 first or at the same time. Folic acid alone can precipitate or worsen subacute combined degeneration of the cord. - Monitor plasma potassium during the initial correction phase in severe deficiency. Rapid haematological response can cause hypokalaemia. Refer any patient who becomes unwell, develops palpitations or muscle weakness during initiation. - People taking chloramphenicol may respond poorly to hydroxocobalamin. Where the response is inadequate, review the medication history and refer. - Serum concentrations may be lowered by oral contraceptives. This interaction is unlikely to be clinically significant but should be considered when interpreting a borderline result. - Breastfeeding is not a contraindication. Hydroxocobalamin is excreted in breast milk but is unlikely to harm the infant. - Inform the patient that urine may appear reddish after the injection. - Anaphylaxis is rare but recognised. Facilities and trained staff for the management of anaphylaxis must be available, with immediate access to adrenaline (epinephrine) 1 in 1,000 injection and a telephone. Observe the patient after administration. - Use with caution where there is a history of significant allergy or asthma.
Actions if patient is excluded or declines treatment	Advise on alternative options and how these can be accessed, including their GP practice. Where referral is required because of neurological symptoms, an abnormal blood picture or pregnancy, make the referral clear and timely and do not delay it by starting treatment. Document any advice given and the decision reached. Inform or refer to the GP as appropriate.

Details of the medicine

Name, form and strength of medicine	Hydroxocobalamin 1mg/ml Solution for Injection
Legal category	POM

Storage	Store below 25°C. Protect from light. Do not freeze.
Route/method of administration	Intramuscular injection. Record the anatomical site used and rotate sites where a course is given.
Dose and frequency of administration	● Initial correction: 1 mg intramuscularly three times a week for 2 weeks. ● Where neurological involvement is suspected the schedule differs and the patient is excluded from this PGD; refer. ● Maintenance where the deficiency is NOT diet related: 1 mg intramuscularly every 2 to 3 months, continuing long term. A daily high dose oral alternative (500 to 1000 micrograms) may be considered as an alternative for a patient who prefers it, in line with CKS. ● Maintenance where the deficiency IS diet related: cyanocobalamin 50 to 150 micrograms daily by mouth (see PGD 2 of 3), or hydroxocobalamin 1 mg intramuscularly twice yearly.
Quantity to be administered and/or supplied	One 1 mg ampoule per administration.
Maximum or minimum treatment period	● Initial correction of 2 weeks, followed by maintenance for as long as clinically indicated. Where the cause is not diet related, maintenance is usually lifelong and may continue under this PGD. ● Review at least annually, including symptoms, response and adherence, and repeat a full blood count and B12 at least annually while on maintenance. ● Repeat blood testing is not required before each maintenance injection once the diagnosis and cause are established. ● Refer to the GP if there is no response to the initial correction, if new neurological symptoms develop at any point, if the blood picture changes, or if the cause of the deficiency has never been established.
Adverse effects	Injection site reactions, dizziness, nausea, headache, rash, and rarely anaphylaxis. Urine may appear reddish after injection. Hypokalaemia may occur during initial correction of severe deficiency. Refer to the SPC for the full list.
Yellow card reporting	Report suspected adverse reactions to the MHRA via the Yellow Card scheme at https://yellowcard.mhra.gov.uk. Document any adverse reaction in the individual's record and inform their GP.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP with whom the individual is registered ● name of healthcare practitioner ● name and brand of medication ● date of supply, dose, form, route and quantity supplied ● batch number, expiry date and anatomical site of administration ● the blood results relied on, including the date they were taken and the laboratory or device used ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken ● that the medicine was supplied or administered via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. The individual's

	GP should be informed. ● Adults aged 18 years and over: keep records for 8 years ● Children aged under 18 years: keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medicine, and a written record of the dose given and the date the next dose is due.
Follow-up advice to be given to patient or carer	● Hydroxocobalamin starts to work straight away, but it may take a few days or weeks before B12 levels and symptoms such as extreme tiredness or lack of energy start to improve. ● At first the injection may be needed a few times a week to build levels up. Once the condition improves it may only be needed every few months. ● There may be some pain, swelling or itching where the injection was given. This is usually mild and does not last long. ● It is safe to take long term. Some people need to continue it for the rest of their lives. ● Urine may look reddish for a short time after the injection, which is harmless. ● Seek medical advice promptly for any new numbness, tingling, unsteadiness, memory or mood change, and urgent advice for any breathing difficulty, facial swelling or widespread rash after an injection.

Patient Group Direction 2 of 3

for the supply of

Cyanocobalamin 50 microgram Tablets

for maintenance of diet related vitamin B12 deficiency

Clinical condition or situation to which this PGD applies

PGD Indication	Oral maintenance of vitamin B12 in adults whose deficiency is diet related, for example a vegan or severely restricted diet, following confirmation of deficiency and where injection is not required or not preferred. This PGD provides a consistent supply and counselling route; cyanocobalamin 50 microgram tablets are a pharmacy medicine and may also be sold over the counter.
Inclusion criteria	• Adults aged 18 years and over. • Confirmed vitamin B12 deficiency assessed as diet related, with malabsorption and pernicious anaemia considered and thought unlikely. • A full blood count, blood film and serum folate have been obtained and reviewed alongside the B12 result. • Valid informed consent obtained.
Exclusion criteria	• Deficiency due to, or possibly due to, malabsorption, pernicious anaemia, gastric or ileal surgery, or any cause other than diet. Oral replacement at this dose is not reliable in malabsorption; use the injection pathway or refer. • New or progressive neurological symptoms or signs. Refer for same-week medical assessment. • Known hypersensitivity to cyanocobalamin or to any excipient. • Leber’s hereditary optic neuropathy, or a family history of it. • Pregnancy. Refer to the GP or midwife. • Abnormal full blood count or blood film beyond macrocytic anaemia. Refer. • Under 18 years of age.
Cautions including any relevant action to be taken	• Where both B12 and folate are low, treat B12 first or at the same time. • Adherence matters: oral maintenance only works if taken daily and continued. Where adherence is likely to be poor, the injection pathway may be more appropriate. • Check that the patient is not already taking a B12 containing supplement, as this can raise measured levels without correcting deficiency.
Actions if patient is excluded or declines treatment	Advise on alternative options, including the injection pathway and their GP practice. Document any advice given and the decision reached. Inform or refer to the GP as appropriate.

Details of the medicine

Name, form and strength of medicine	Cyanocobalamin 50 microgram tablets
Legal category	P
Storage	Store below 25°C in the original container, protected from light.
Route/method of administration	Oral.
Dose and frequency of administration	For diet related deficiency: 50 to 150 micrograms daily, taken between meals. Where a higher oral dose is considered appropriate for non diet related deficiency in line with CKS (500 to 1000 micrograms daily), this

	is outside this PGD; discuss with the GP.
Quantity to be administered and/or supplied	Up to 3 months supply at the selected dose, with review before further supply.
Maximum or minimum treatment period	Continue for as long as the dietary cause persists. Review at least annually with symptoms, adherence and a repeat B12 and full blood count. Refer to the GP if symptoms persist or recur despite adherence, or if a non dietary cause becomes likely.
Adverse effects	Uncommon. Nausea, headache, dizziness and rash have been reported. Refer to the SPC for the full list.
Yellow card reporting	Report suspected adverse reactions to the MHRA via the Yellow Card scheme at https://yellowcard.mhra.gov.uk. Document any adverse reaction in the individual's record and inform their GP.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP with whom the individual is registered ● name of healthcare practitioner ● name and brand of medication ● date of supply, dose, form, route and quantity supplied ● batch number and expiry date of the pack supplied ● the blood results relied on, including the date they were taken and the laboratory or device used ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken ● that the medicine was supplied or administered via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. The individual's GP should be informed. ● Adults aged 18 years and over: keep records for 8 years ● Children aged under 18 years: keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medicine, together with written dietary advice on B12 sources.
Follow-up advice to be given to patient or carer	Take the tablets every day, between meals. Improvement in tiredness and other symptoms may take a few weeks. Continue while the dietary cause persists and attend for annual review. Seek medical advice promptly for any new numbness, tingling, unsteadiness, or memory or mood change.

Patient Group Direction 3 of 3

for the supply of

Folic acid 5mg Tablets

for the treatment of folate deficiency

Clinical condition or situation to which this PGD applies

PGD Indication	Treatment of confirmed folate deficiency in adults identified on blood testing, where vitamin B12 deficiency has been excluded or is being treated at the same time.
Inclusion criteria	• Adults aged 18 years and over. • Serum folate below 7 nanomol/L (3 micrograms/L), or 7 to 10 nanomol/L with supporting clinical features, interpreted with the full blood count and blood film. • Vitamin B12 deficiency has been excluded, OR B12 deficiency is present and treatment with hydroxocobalamin has been started first or at the same time under PGD 1 of 3. • Valid informed consent obtained.
Exclusion criteria	• Vitamin B12 status unknown, or B12 deficiency present and untreated. Folic acid alone can precipitate or worsen subacute combined degeneration of the spinal cord. Test and treat B12 first, or refer. • Known hypersensitivity to folic acid or to any excipient. • Known or suspected malignancy, unless folate deficiency has been confirmed and the treating specialist agrees, because folic acid may accelerate the progression of some tumours. • Pregnancy or planning pregnancy. Folic acid dosing in pregnancy, including the 5 mg dose for higher risk groups, belongs with the GP or midwife. Refer. • Taking methotrexate, phenytoin, phenobarbital, primidone or another antifolate or antiepileptic where folate supplementation may alter treatment. Refer to the GP. • Unexplained anaemia where the cause has not been established, or an abnormal blood picture beyond macrocytic anaemia. Refer. • Under 18 years of age.
Cautions including any relevant action to be taken	• Always confirm B12 status before starting folic acid. This is the single most important safety check in this PGD. • Establish the cause of the folate deficiency. Poor diet, alcohol excess, malabsorption including coeliac disease, pregnancy, haemolysis and certain medicines are common causes; where the cause is unclear or suggests malabsorption, refer. • Where anaemia does not correct after a course, refer rather than repeating treatment.
Actions if patient is excluded or declines treatment	Advise on alternative options and how these can be accessed, including their GP practice, and give dietary advice. Document any advice given and the decision reached. Inform or refer to the GP as appropriate.

Details of the medicine

Name, form and strength of medicine	Folic acid 5mg tablets
Legal category	POM
Storage	Store below 25°C in the original container, protected from light.
Route/method of administration	Oral.

Dose and frequency of administration	Folic acid 5 mg once daily. In most people treatment is required for 4 months.
Quantity to be administered and/or supplied	Up to 4 months supply, or a shorter supply with planned review.
Maximum or minimum treatment period	Usually 4 months. Where the underlying cause persists, for example malabsorption, longer treatment may be needed and the patient should be referred to the GP rather than continued indefinitely under this PGD. Repeat full blood count and folate at the end of the course, and refer if not resolved.
Adverse effects	Uncommon. Nausea, loss of appetite, bloating and rash have been reported. Refer to the SPC for the full list.
Yellow card reporting	Report suspected adverse reactions to the MHRA via the Yellow Card scheme at https://yellowcard.mhra.gov.uk. Document any adverse reaction in the individual's record and inform their GP.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP with whom the individual is registered ● name of healthcare practitioner ● name and brand of medication ● date of supply, dose, form, route and quantity supplied ● batch number and expiry date of the pack supplied ● confirmation that B12 status was checked and the result relied on ● the blood results relied on, including the date they were taken and the laboratory or device used ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken ● that the medicine was supplied or administered via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. The individual's GP should be informed. ● Adults aged 18 years and over: keep records for 8 years ● Children aged under 18 years: keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medicine, together with written dietary advice.
Follow-up advice to be given to patient or carer	Take one tablet daily for the full course, usually 4 months. Good dietary sources of folate include broccoli, Brussels sprouts, asparagus, peas, chickpeas and brown rice. Attend for a repeat blood test at the end of the course. Seek medical advice if symptoms persist or worsen, and promptly for any new numbness, tingling or unsteadiness.

Key references

- NICE CKS: Anaemia, vitamin B12 and folate deficiency
- NICE NG239: Vitamin B12 deficiency in over 16s, diagnosis and management
- Summary of Product Characteristics for Hydroxocobalamin 1mg/ml solution for injection, Cyanocobalamin 50 microgram tablets and Folic acid 5mg tablets, current versions
- British Society for Haematology guidance on the diagnosis and treatment of cobalamin and folate

disorders

- British National Formulary (BNF)
- NICE Medicines Practice Guideline 2 (MPG2): Patient Group Directions
- NICE MPG2 competency framework for health professionals using patient group directions

Agreement to practise by the Registered Healthcare Professional

By signing this PGD you are confirming that you agree to its contents and that you will work within it. PGDs do not remove inherent professional obligations or accountability. It is the responsibility of each healthcare professional to practise only within the bounds of their own competence and professional code of conduct.

Name and address of pharmacy premises / clinic premises to which this PGD relates	

Name of healthcare professional	Job title	Registration number	Signature

Valid from date	Expiry date

PGD adoption and authorisation by the employer/clinical lead

The employer has ensured that the person using this PGD, has the knowledge and skills to safely provide the service which will encompass the supply or administration of a medication or vaccine.

This section must be signed by the superintendent or clinical lead responsible for this service.

To be completed by the adopting pharmacy: the first block is signed by that pharmacy's superintendent or clinical lead, and the second by its professional group signatory. Get Real Health signs the authorship block below, not these.

Name of superintendent / clinical lead	Job title & name of organisation	Signature	Date
Name of professional group signatory	Job title & name of organisation	Signature	Date

Signed on behalf of Get Real Health

Name	Qualification	Signature	Date
Nitin Shori	Medical Doctor GMC: 6047293		30/07/2026
Chris Pilkington	Pharmacist GPhC: 2046322		30/07/2026

Change history

Version number	Change details	Date
001	Development and issue of new PGD, hydroxocobalamin injection	25th August 2025
002	Clinical review (J. Wilkins with C. Pilkington): investigation set, B12 interpretation thresholds with active B12 preferred, and pregnancy and breastfeeding wording added. Signatory details corrected.	29th July 2026
003	Expanded to three medicines following PPH clinical review. Full blood interpretation section added covering full blood count, blood film, B12 and folate with NICE thresholds, limitations and the requirement to treat B12 before or alongside folate. Breastfeeding changed from excluded to included, pregnancy remains excluded with the reason stated. Cautions added for chloramphenicol and oral contraceptives. Medicine name corrected to Hydroxocobalamin 1mg/ml Solution for Injection. Maintenance dosing split between diet related and non diet related causes, with lifelong maintenance permitted under the PGD subject to annual review and annual bloods. Records now require batch number, expiry date, site of administration and the blood results relied on. Patient follow-up advice expanded. Cyanocobalamin 50 microgram tablets added as PGD 2 of 3 for diet related maintenance, and Folic acid 5mg tablets added as PGD 3 of 3 for folate deficiency. Point of care B12 only devices stated as insufficient on their own to start treatment.	30th July 2026
004	Adoption and authorisation blocks corrected: those two signatures belong to the adopting pharmacy’s superintendent and professional group signatory, not to Get Real Health, and had been pre-filled in error. Clinical content unchanged.	6th August 2026

Version and change record
Version: v005
Issued: 10 September 2026
Supersedes: Version 004, 6 August 2026
Change: the summary of the governing guidance is added as part 2 of the document, between the cover and the PGD, which is the house format for every Get Real Health PGD. It is summarised from the source it names and adds nothing to it. Nothing else in this document is altered, and this is not a clinical review of the remainder.

Version and change record
Version: v006
Issued: 10 September 2026
Supersedes: Version 005, 10 September 2026
Valid from: 10 September 2026. Expiry: 31 July 2027. These dates govern this version and supersede any earlier date printed elsewhere in this document.
Changes in this version:
Validity: this version is valid from 10 September 2026 and expires on 31 July 2027. The signed master previously stated no period of validity, or an expiry of 31 October 2026; those are superseded. No expiry or review date was stated.

Signed on behalf of Get Real Health

Version v006, 10 September 2026.

Name	Qualification	Signature	Date
Nitin Shori	Medical Director GMC: 6047293		10 September 2026
Chris Pilkington	Head Pharmacist GPhC: 2046322		10 September 2026

Both authorising signatories reviewed this version together on 10 September 2026 and gave their authorisation for it to be issued. Signatures were applied digitally on their joint instruction, which is the established process for Get Real Health PGDs. This version is not valid without both signatures.